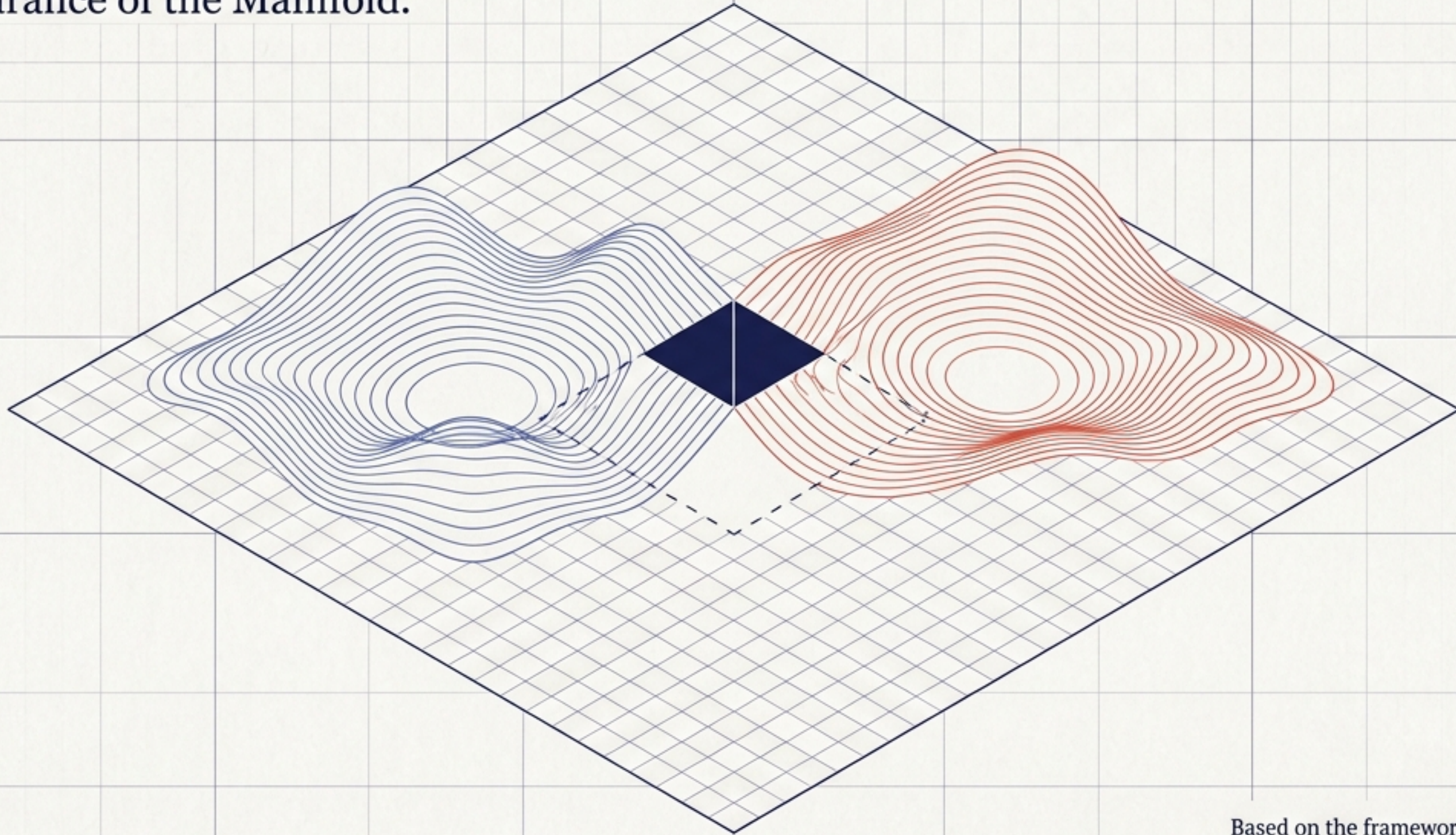


WHERE FIELDS MEET

Dressing, Relational Invariance, and
the Disappearance of the Manifold.

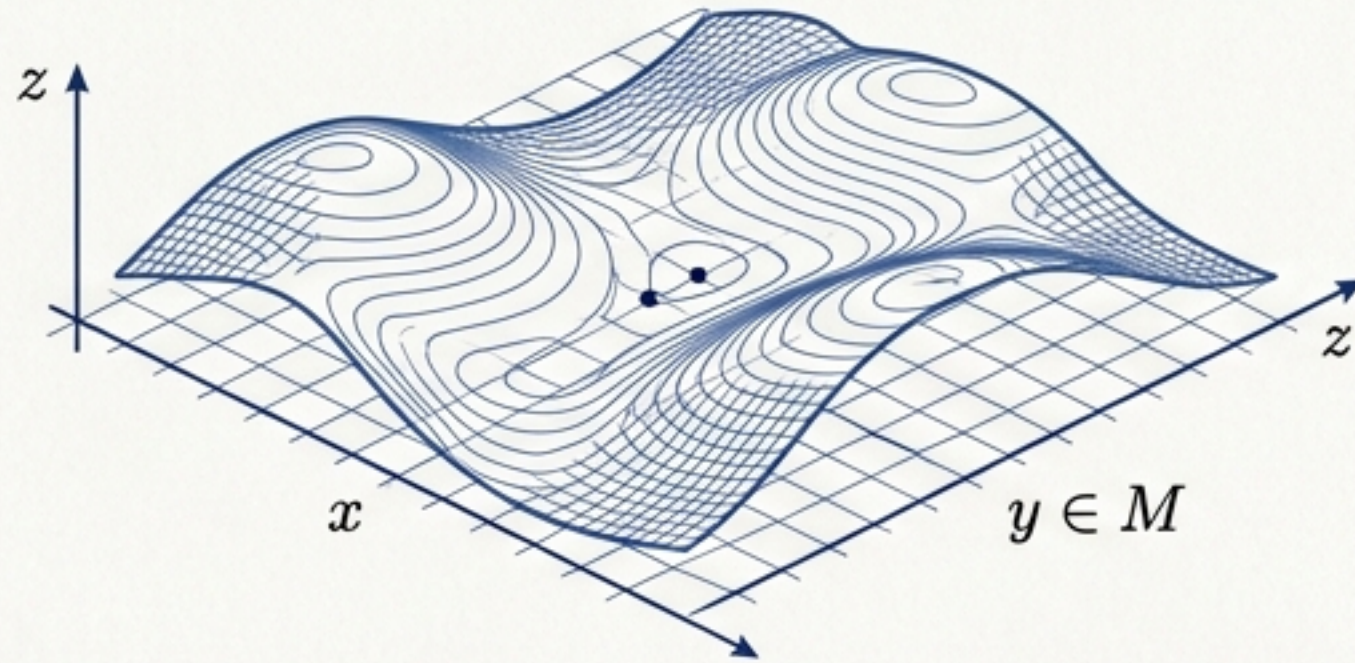
Distilling the transition
from general covariance
to manifest relationalism
in quantum field theory.



Based on the frameworks of J. François, L. Ravera,
and the theoretical proposals of Flyxion (2026).

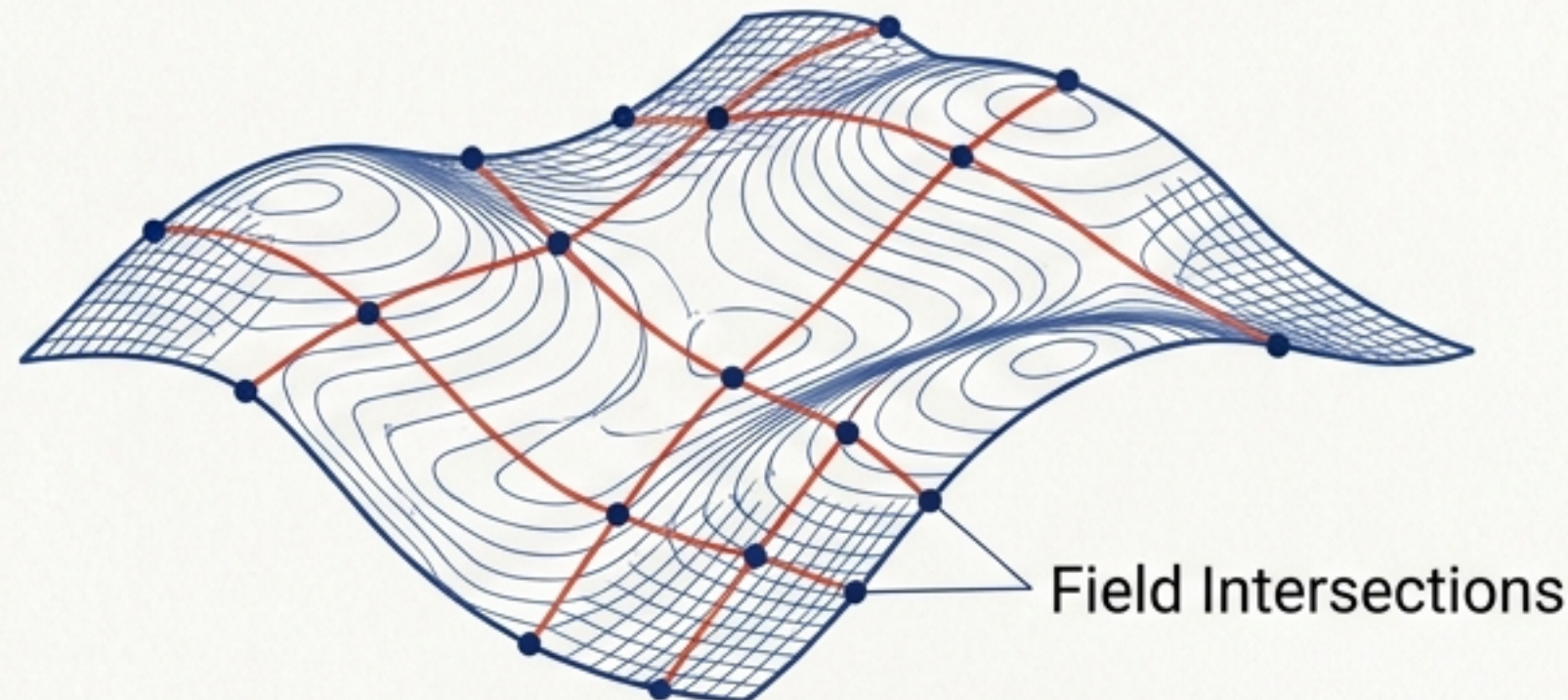
THE MANIFOLD IS AN INDISPENSABLE SCAFFOLD WITH NO PHYSICAL CONTENT

THE MATHEMATICAL PRESENTATION



General relativity and the Standard Model are written on a differentiable manifold. But physically, spacetime is simply where fields meet.

THE PHYSICAL REALITY



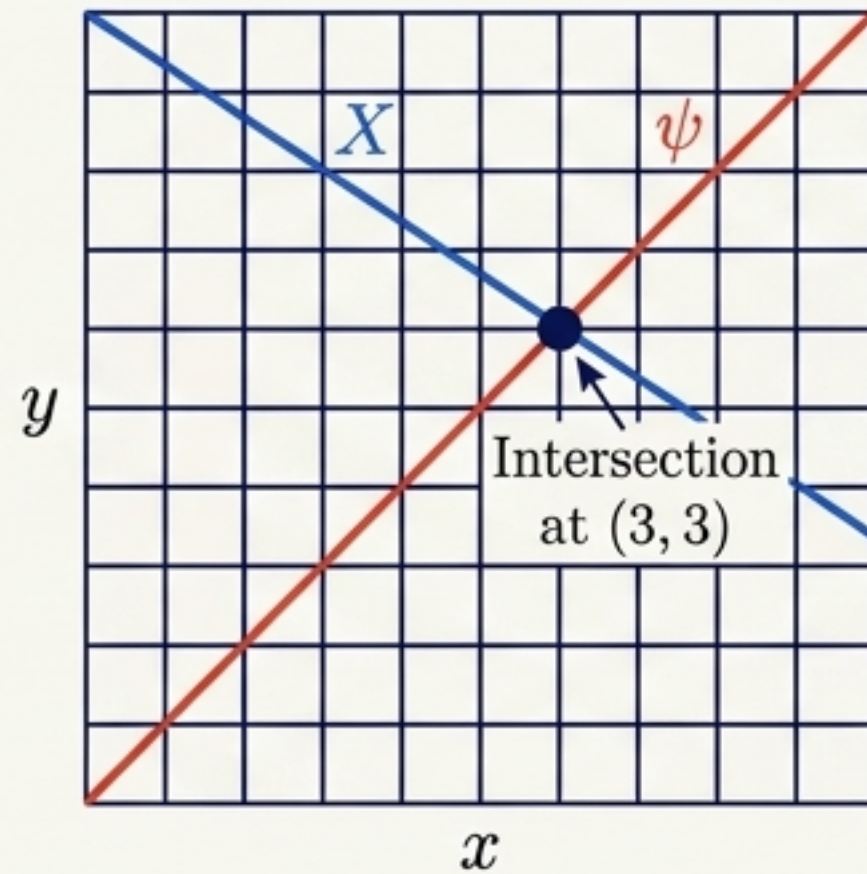
$$\text{Let } S[\phi] = \int_M L(\phi, d\phi, \dots).$$

Under a local symmetry group G , $\phi \neq \phi^\gamma$ in field space $\mathcal{F}$, but $[\phi]_G = [\phi^\gamma]_G$ in the orbit space $\mathcal{F}/G$.

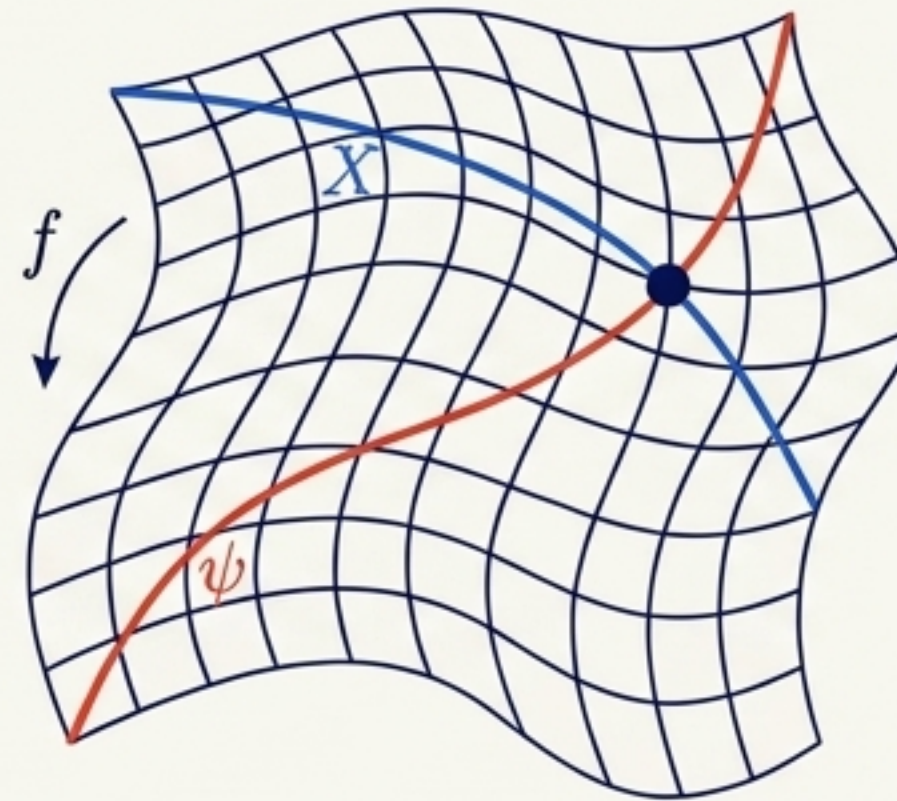
Takeaway: The formalism distinguishes what the dynamics cannot.

Coordinate Labels Carry No Physical Consequence

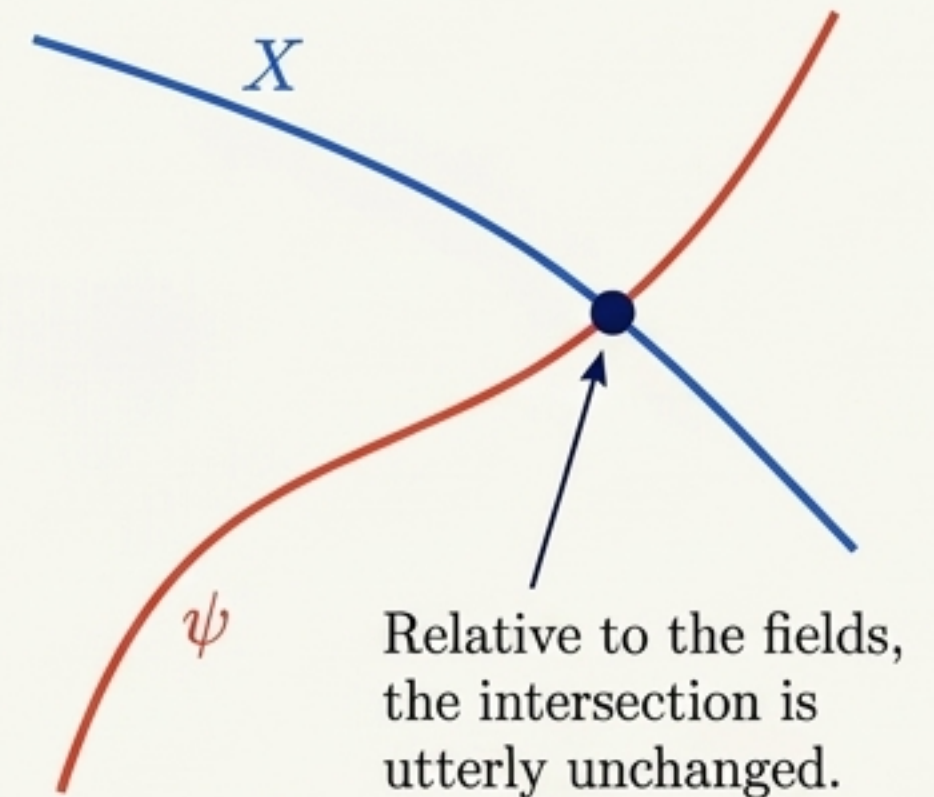
Step 1: Bare Coordinates



Step 2: Active Diffeomorphism



Step 3: Relational Invariance



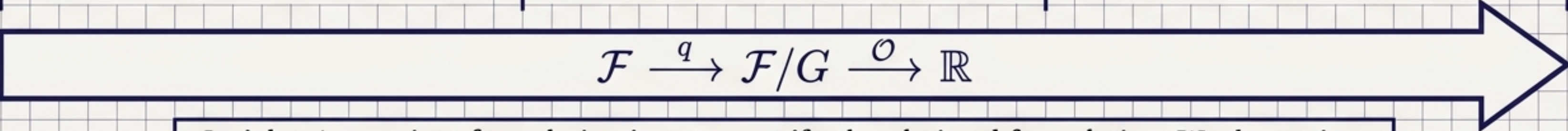
The Relational Observable: $\Psi(\xi) = \psi(X^{-1}(\xi))$

Physical Event $\neq x \in M$ | Physical Event = [Coinciding field values and their relations]

Takeaway: An entity is not a static substrate. It is a distinction maintained through transformations.

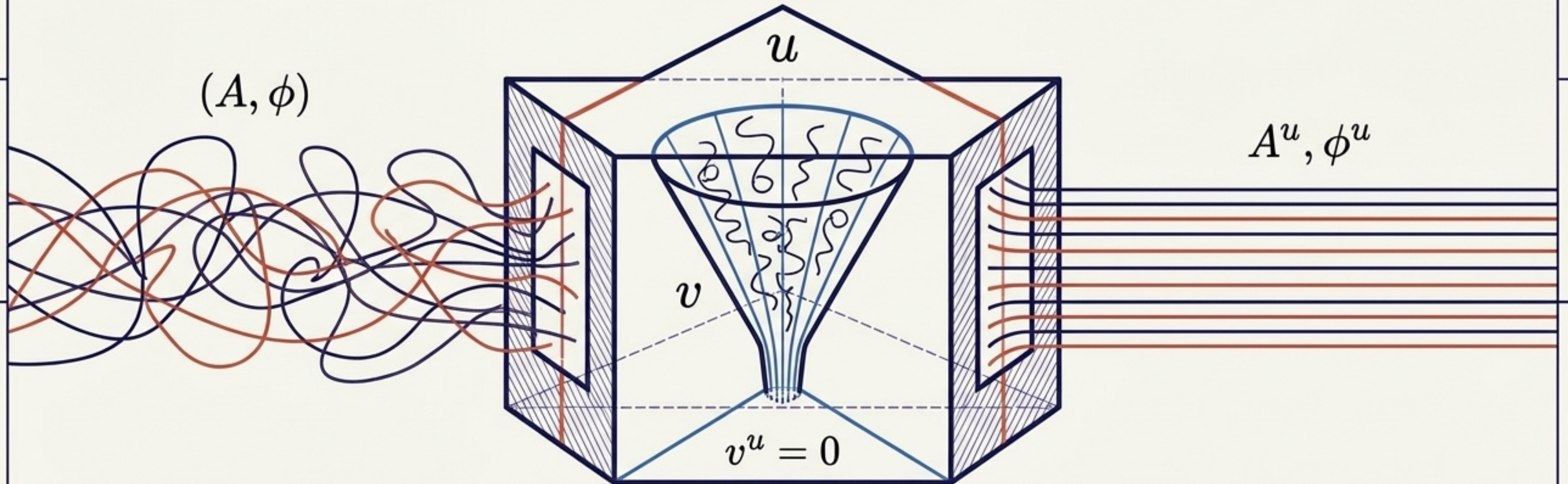
General Covariance is Not Manifest Relationalism

Encoded (Tacit Relationalism)	Recoverable (Complete Observables)	Manifest (Natural Availability)
 Standard formulation. Compute on $\mathcal{F}$, establish invariance afterward.	 Invariant maps are extracted via constraint solving or complete observables.	 Variables natively live on the reduced physical space $\mathcal{F}/G$.



Insight: A covariant formulation is not a manifestly relational formulation. We do not just want a theory that hides its relationalism; we want one built entirely out of it.

Constructing Invariance from the Inside



The Rule

DFM extracts invariant variables by drawing a dressing field u from the theory's own field content to compensate for local symmetry. $u^\gamma = \gamma^{-1}u$.

The Composites

$$A^u = u^{-1}Au + u^{-1}du.$$

The curvature dresses perfectly to

$$F^u = u^{-1}Fu.$$

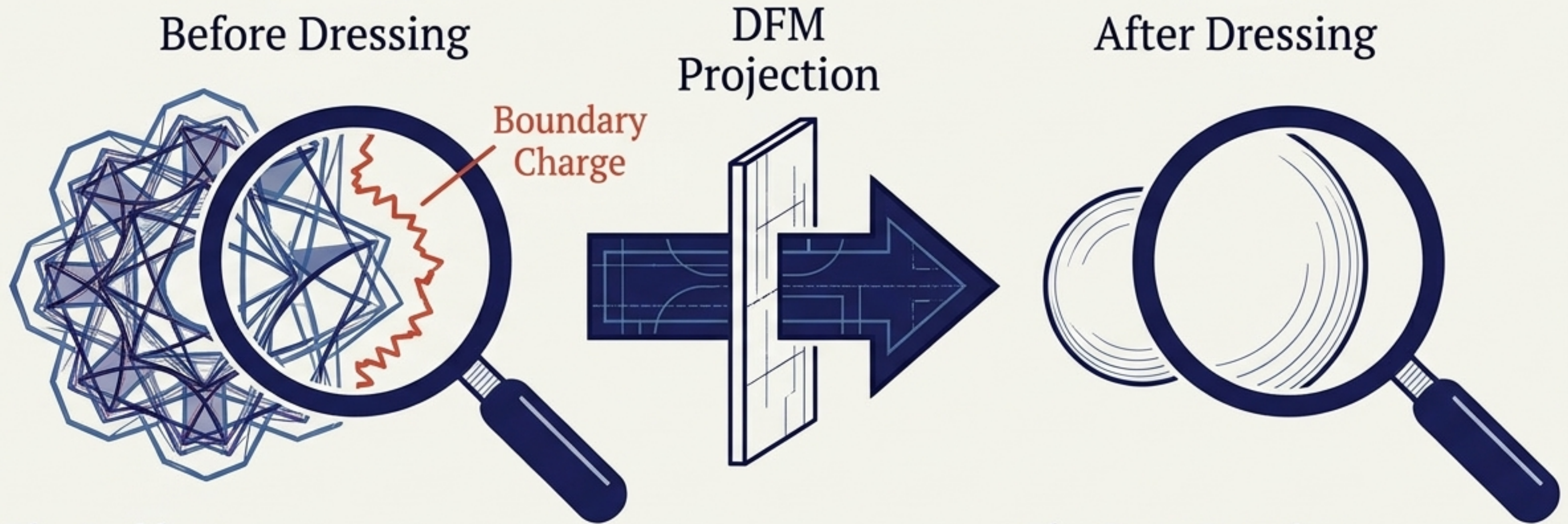
Takeaway

This is not gauge fixing. It is a rewriting of the physical recipe using only internal ingredients.

Evaluating the Architecture of Reduction Procedures

	Output	Global Status	Relational Reading	Ghost Treatment
Gauge Fixing	Section of $\mathcal{F} \rightarrow \mathcal{F}/G$	Obstructed by Singer	None	Track redundancy
Quotienting	Points of $\mathcal{F}/G$	Not a manifold	Implicit	N/A
BRST	Cohomology of s	Inherits gauge issues	Cohomological	Used dynamically
Dressing	Invariant composites ϕ^u	Transition data k_{ij} invariant	Manifest	$v^u = 0$ (They disappear)

Invariance Tells Us What Survives, Not If It Was Safe



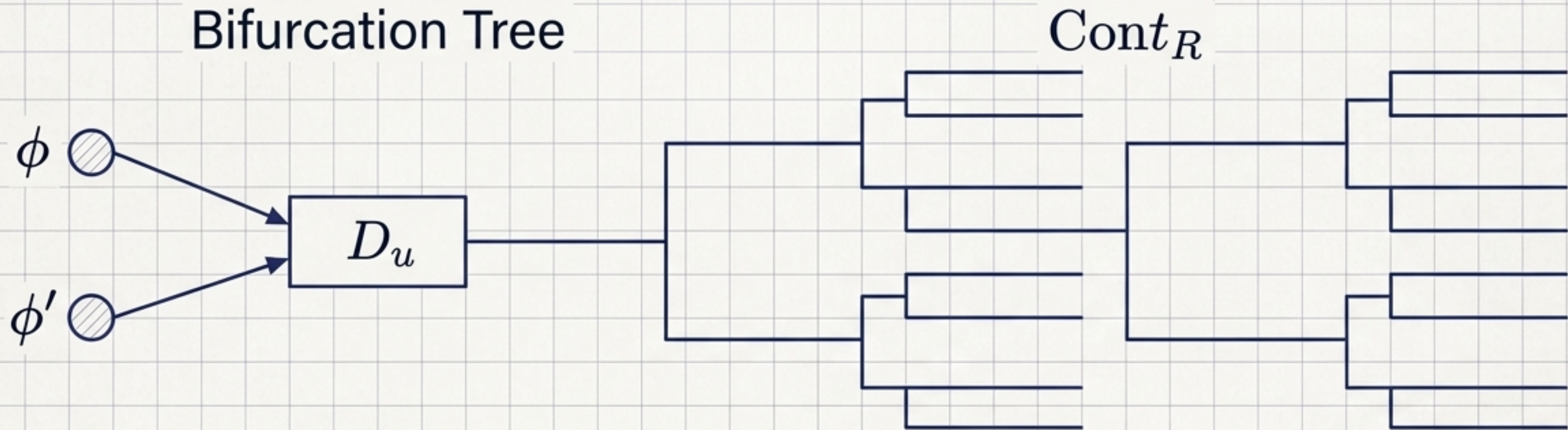
The Problem:

A dressing removes distinctions. But a representational difference is physically consequential if it survives an admissible operational projection (e.g., boundaries, subsystem cuts).

The Open Question:

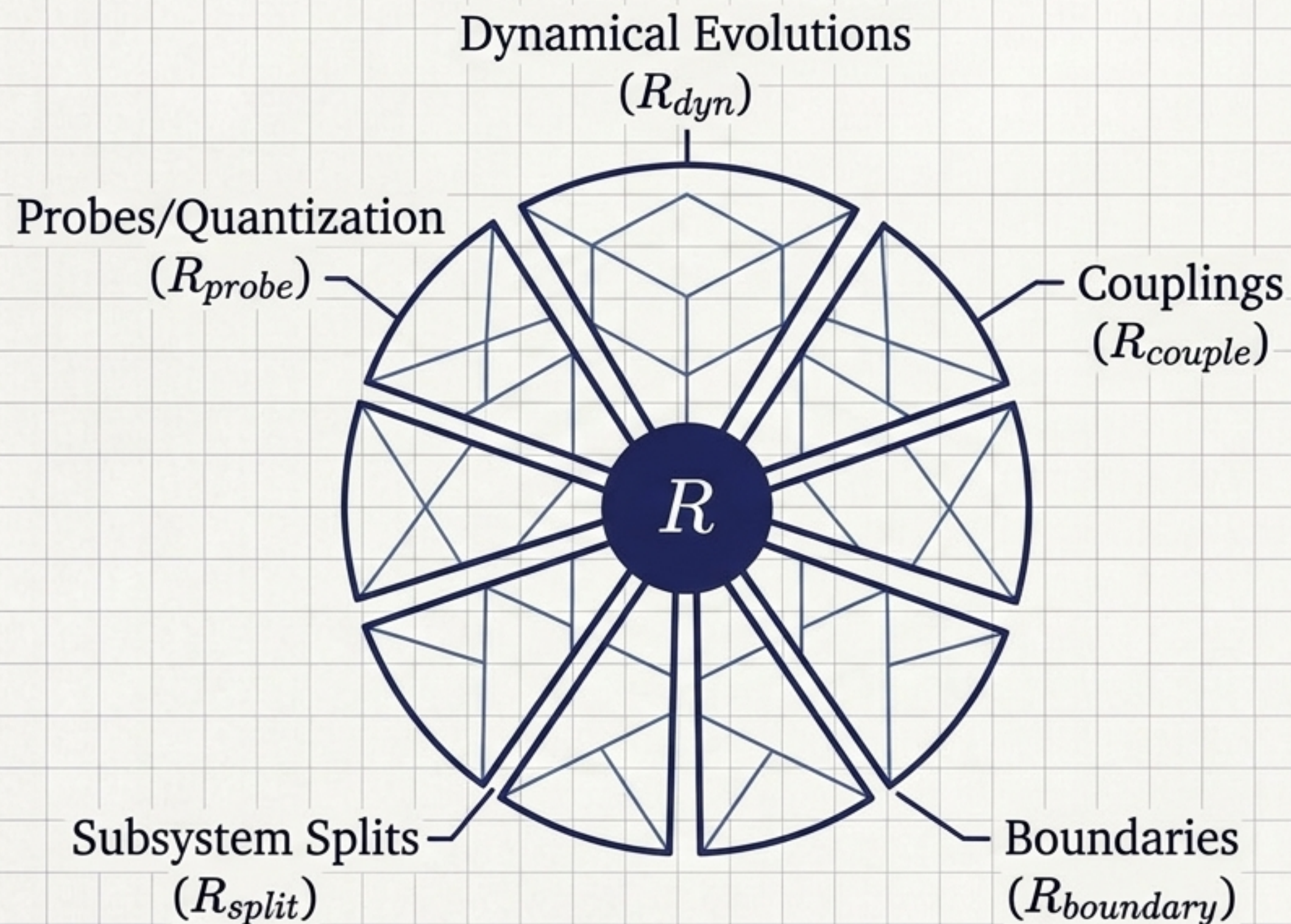
What must an operation preserve for the resulting description to count as physically sufficient?

Dressing as an Admissible Interface



- **The Proposal:** A dressing is adequate only if configurations it identifies as the same support the exact same admissible continuations under physical operations.
- **The Interface Equation:** $D_u(\phi) = D_u(\phi') \Rightarrow Cont_R(\phi) \simeq Cont_R(\phi')$
- **Shift in Perspective:** Replace 'do they look the same now?' with 'can they do the same things in the future?'

Defining the Operational Signature



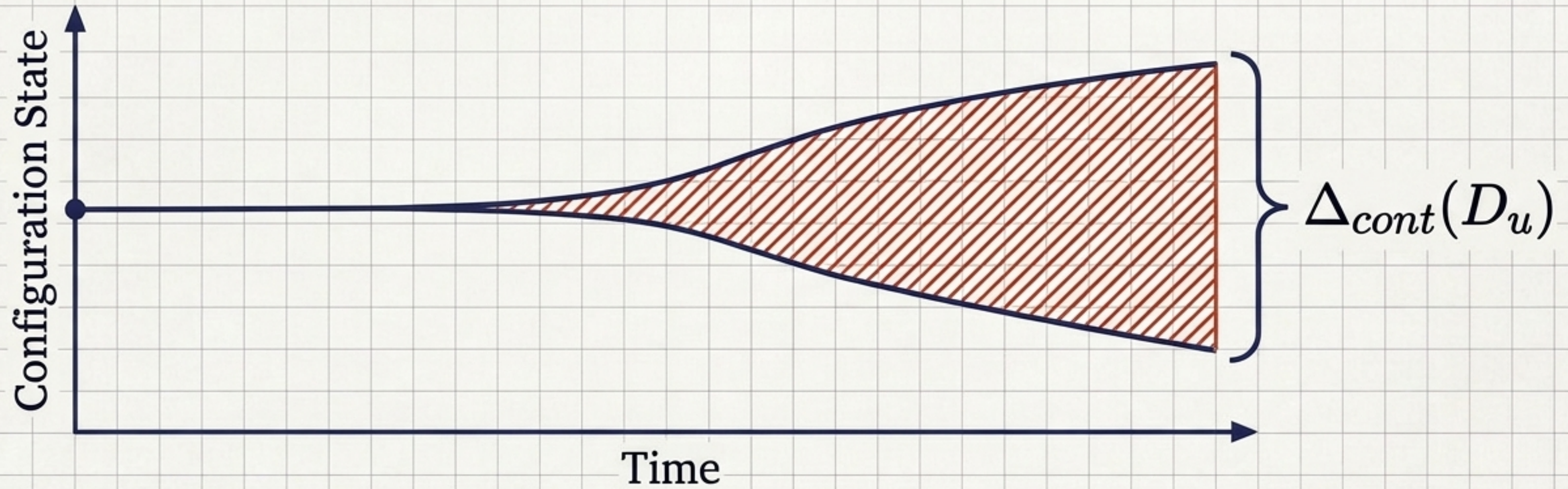
The Monotonicity Law:

$$R \subseteq R' \Rightarrow Adeq_{R'}(D_u) \Rightarrow Adeq_R(D_u)$$

Insight:

Adequacy is entirely relative. A dressing that is adequate for a closed system may fail instantly when you introduce a subsystem cut because it erases vital boundary labels.

Turning Adequacy into a Computable Quantity



Concept:

Rather than a binary pass/fail, we measure the “**continuation loss**” of a dressing map.

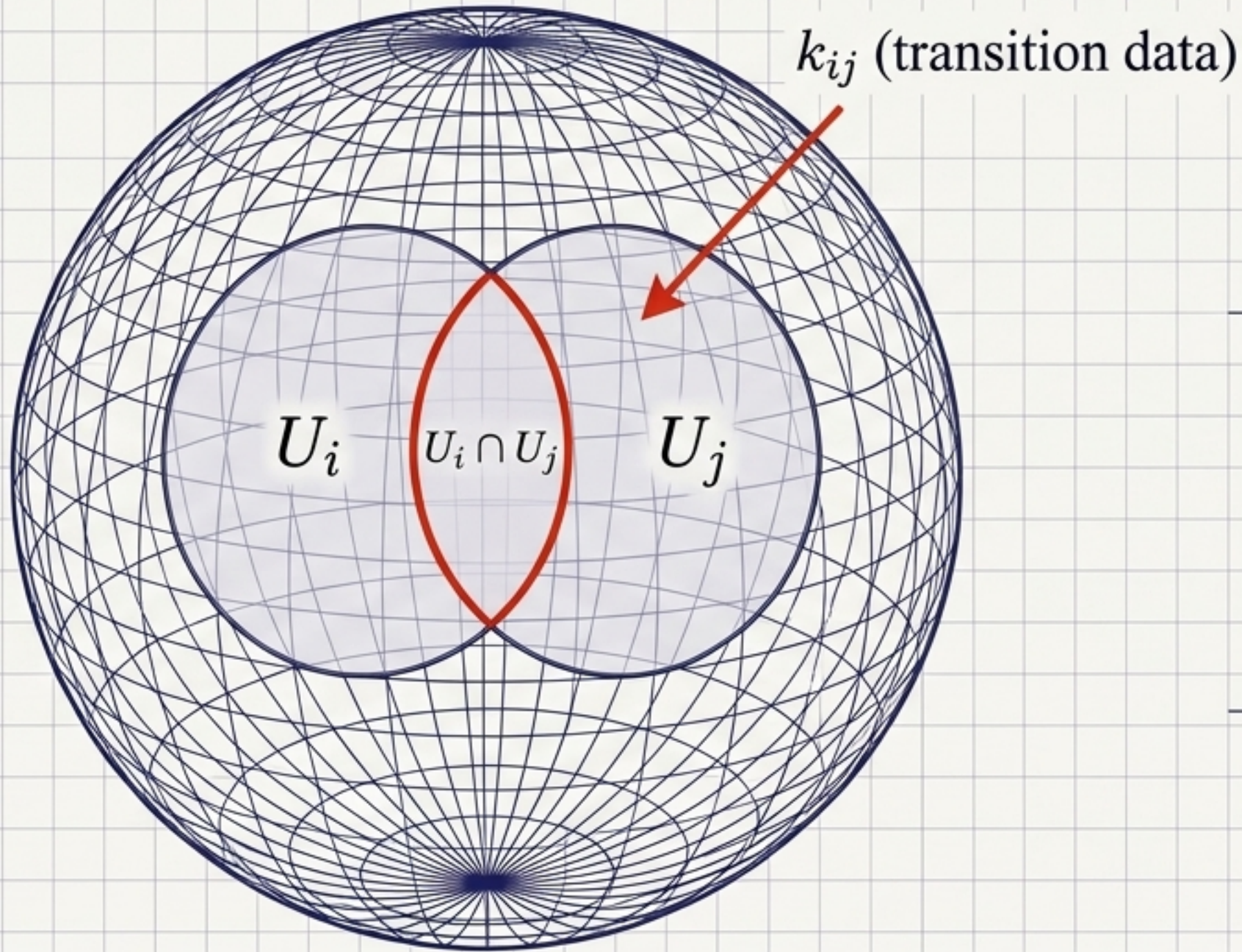
The Loss Functional:

$$\Delta_{cont}(D_u) = \sup_{D_u(\phi)=D_u(\phi')} d_{cont}(Cont_R(\phi), Cont_R(\phi'))$$

Continuation Completeness:

If equality of dressed states induces a complete step-by-step match of future weights and outcomes (bisimulation), the loss is perfectly zero.

The Inescapability of Global Obstructions

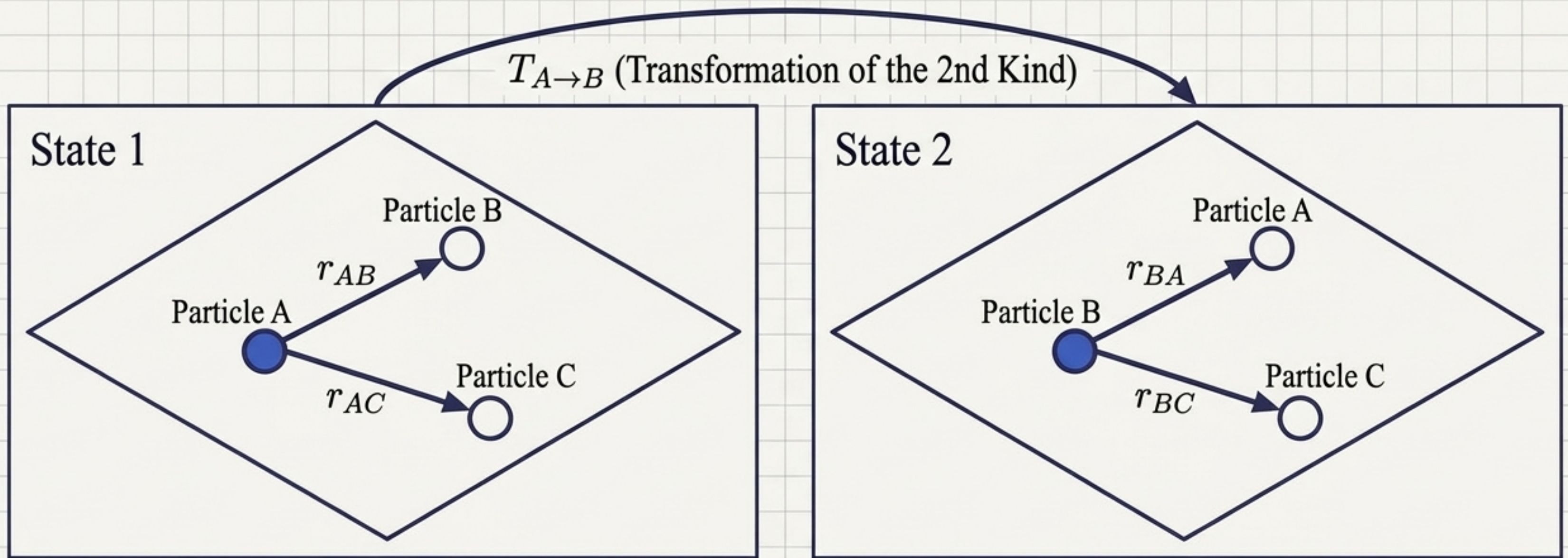


Gribov-Singer Limits: DFM circumvents local Gribov copies, but does not solve the topological impossibility of a perfect global gauge (Singer's theorem). No smooth, global group-valued field-dependent dressing exists.

Relational Holonomy: DFM relocates the obstruction. The transition data ($k_{ij} = u_i^{-1}u_j$) between local patches are gauge-invariant.

Insight: Topological “glitches” are physical data representing residual relational curvature.

Frame Covariance & Transformations of the Second Kind



Relational QM:

In a dressed description, we do not assign an absolute state to a particle. We assign it relative to the rest of the system.

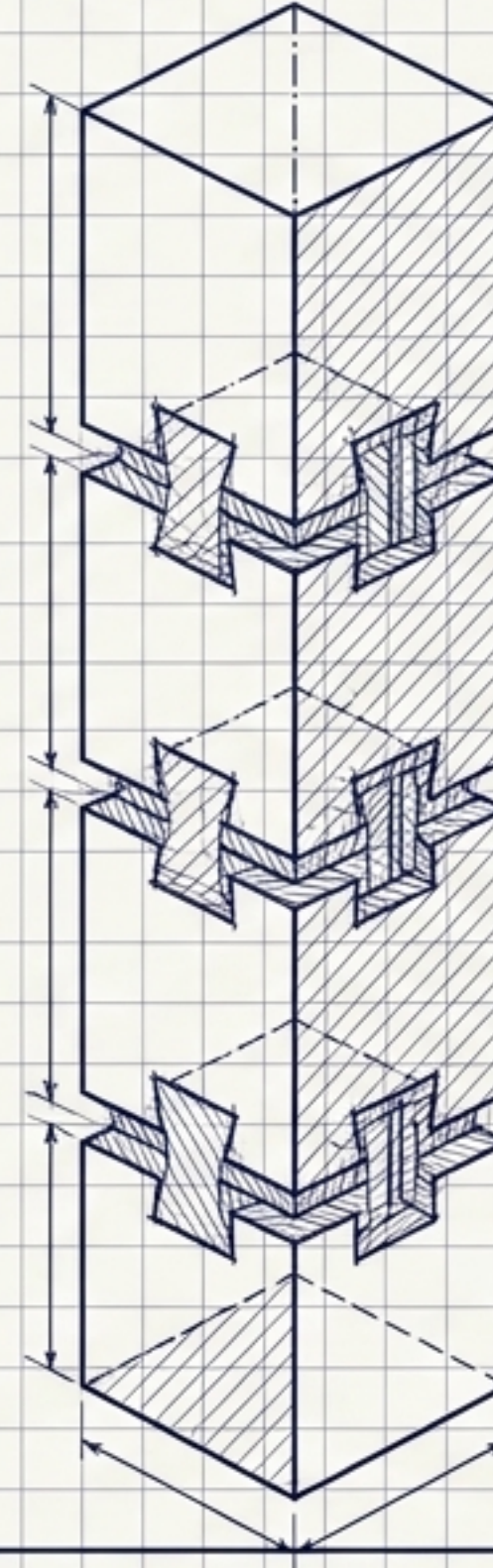
Residual Transformations:

Changing the reference particle ($u' = u\xi$) isn't a gauge redundancy. It is a physical change of relational frame, linking DFM to the Quantum Reference Frame program.

The Blueprint for a Relational QFT

The Target Tuple:

$$\mathcal{Q}_{rel} = (\mathcal{F}_u, \mathcal{A}_u, \omega_u, T_u, R_u)$$



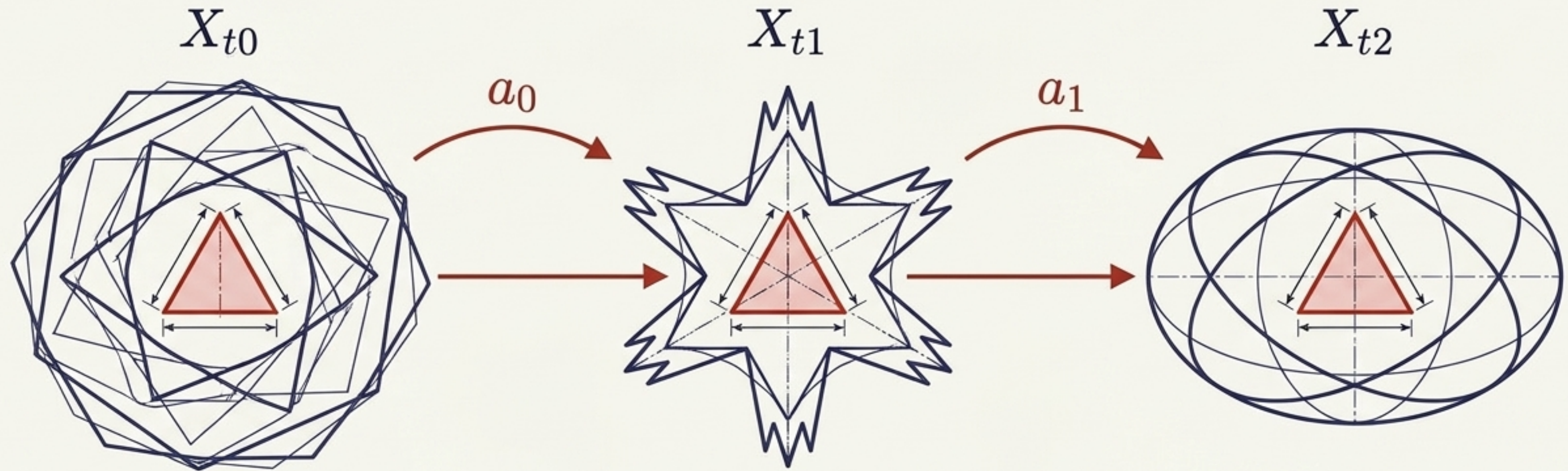
4. **Continuation Completeness:**
Quantization preserves all physical sectors without loss ($\mathcal{C}_{phys} \simeq \mathcal{C}_u$).

3. **Covariant Dynamics:**
Physical evolution intertwines perfectly with frame changes.

2. **Coherent Frame Changes:**
Transformations must compose logically
($T_{b \rightarrow c} \circ T_{a \rightarrow b} = \Omega_{abc} \circ T_{a \rightarrow c}$),
recording residual curvature.

1. **Invariance:** Variables must be strictly invariant under removed redundancies.

Structural Realism Without Empty Relations



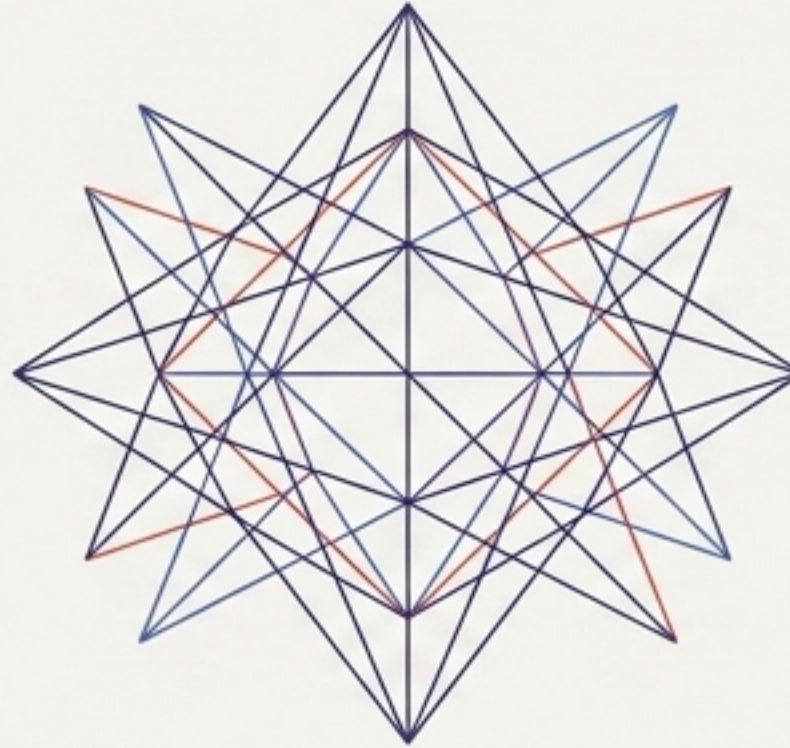
The Philosophical Trap:
Taking away the manifold
does not leave us with
“relations without relata.”

The Redefinition:

Objects are not independent substances.
Objects are relatively stable nodes
of maintained distinction within
structures of admissible continuation.

Identity: Identity equals
a compatible history
of transitions, not numerical
sameness at a coordinate point.

What Remains When the Scaffold is Removed



**Admissible Reduction = Distinction
Collapse + Continuation Preservation**

A successful dressing does more than hide redundancy. It proves that the distinctions it removed were completely unnecessary to the continuation of the physical world.